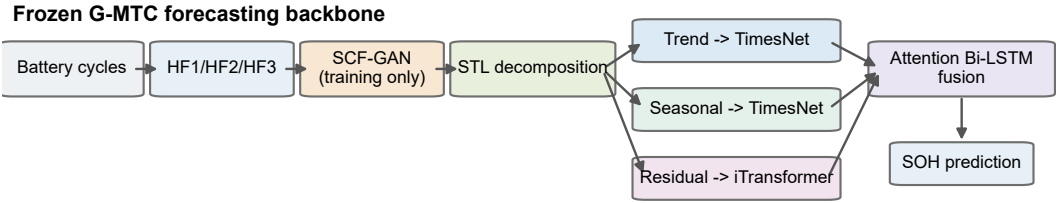
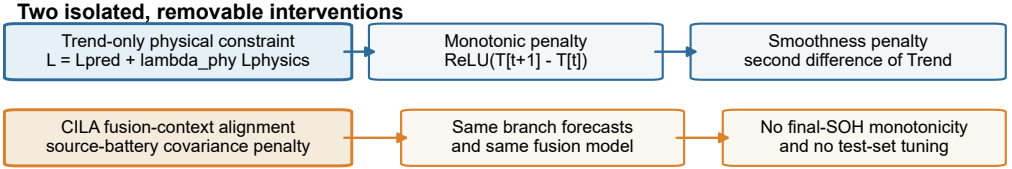
a


Frozen G-MTC forecasting backbone
Battery cycles
HF1/HF2/HF3
SCF-GAN
(training only)
STL decomposition
Trend -> TimesNet
Seasonal -> TimesNet
Residual -> iTransformer
Attention Bi-LSTM
fusion
SOH prediction

b


Two isolated, removable interventions
Trend-only physical constraint
L = Lpred + lambda_phy Lphysics
Monotonic penalty
ReLU(T[t+1] - T[t])
Smoothness penalty
second difference of Trend
CILA fusion-context alignment
source-battery covariance penalty
Same branch forecasts
and same fusion model
No final-SOH monotonicity
and no test-set tuning